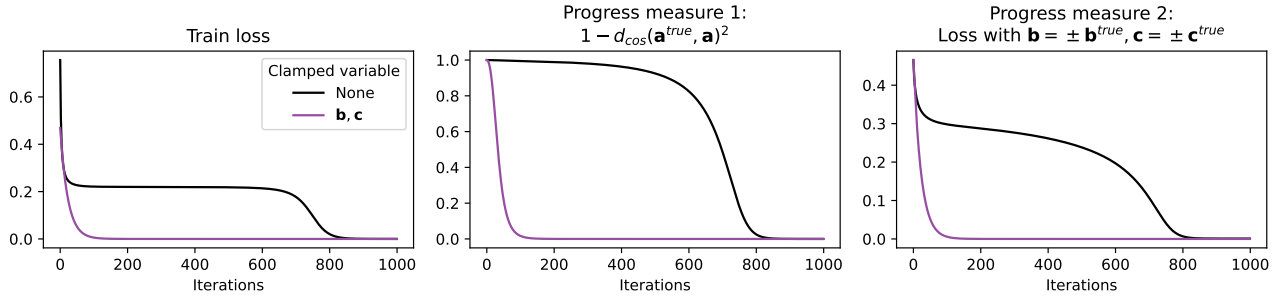


*Figure 9.* Progress measures vs clamping in a toy model. Black curve indicates standard training, as in Figure 5a. Leftmost plot shows the training loss. Right two plots show two different progress measures for the learning of $a$. We note that clamping (purple curve, leftmost plot) gives a clearer signal of the relevant subcircuit than progress measures in standard training (black curves, right two plots).

may consider, which offers a direct analog to clamping, is the loss if $\mathbf{b} = \mathbf{b}^{true}, \mathbf{c} = \mathbf{c}^{true}$.[21] Note the difference between this progress measure and clamping is that with the progress measure, we're considering the evolution of $\mathbf{a}$ during normal training and setting $\mathbf{b} = \mathbf{b}^{true}, \mathbf{c} = \mathbf{c}^{true}$ at every checkpoint. With clamping, we set $\mathbf{b} = \mathbf{b}^{true}, \mathbf{c} = \mathbf{c}^{true}$ throughout training and then consider the evolution of $\mathbf{a}$.

Results are shown in Figure 9. We find that both progress measures considered, cosine distance to true value and loss assuming $\mathbf{b}, \mathbf{c}$ fixed to those at the end of training, exhibit a phase change when observed in isolation (black curves, right two columns, Figure 9). On the other hand, as seen in Figure 5a and reproduced in the leftmost plot of Figure 9 in purple, the loss curve when training with clamping is smoothly exponential and easily interpretable.

We extend results from the toy model to our setting, and consider progress measures corresponding to two of our clamped experiments: that of clamping the previous token attention pattern (clamp Step 1, orange curve, Figure 5) and that of clamping the whole match operation (clamp Steps 1, 2, 3qk, and 4, green curve, Figure 5). We construct analogous progress measures by perfecting either the previous token head attention pattern or induction head attention pattern for each checkpoint in standard training and considering the loss. Figure 10 shows the results, with a reproduction of the relevant curves from Figure 5 for a side-by-side comparison. We can see that the progress measure is informative in the sense that it identifies the copy subcircuit (Steps 3v, 5) as quicker to learn than the subcircuit consisting of Steps 2, 3qkv, 4, and 5. However, when just looking at progress measures, both subcircuits appear to still emerge in a phase change, missing out on the crucial distinction between the two: the copy subcircuit (Steps 3v, 5) does not contain interacting components, whereas the other subcircuit (Steps 2, 3qkv, 4, and 5) does. These results show the benefit of the clamping approach for underlying exponentially forming subcircuits whose dynamics of formation causally affect the learning dynamics of the full network.

To conclude, we discuss pros and cons of progress measures versus our clamping approach. Progress measures have the benefit of only requiring access to network checkpoints after training, reducing the load of the mechanistic interpretability researcher. For cases of large language models, where training data is often not disclosed even for open-source models (Touvron et al., 2023), this could be especially appealing. Furthermore, as each progress measure only involves forward passes on checkpoints, they may be easier to iterate on. On the other hand, progress measures, as demonstrated above, provide largely a correlational understanding of subcircuits giving rise to dynamical phenomena. Clamping experiments, though more complex, provide a more causal understanding of learning dynamics. We hope that our results and open-source artificial optogenetics framework encourages more researchers to consider these types of experiments.

In terms of scaling to large language models, the hope would be that isolated subcircuits evolve quickly enough that the cost is low enough. Specifically, one can view clamping experiments as requiring about 3x the cost of progress measures per iteration (due to the forwards and backwards passes). Thus, if the number of iterations needed to train clamped networks is substantially lower than that of the full network, clamping experiments may actually be cheaper than progress measures.

[21] Specifically, we consider $\mathbf{b} = \pm\mathbf{b}^{true}, \mathbf{c} = \pm\mathbf{c}^{true}$ and pick the one that yields lower loss (this choice is made to avoid the scaling issue).

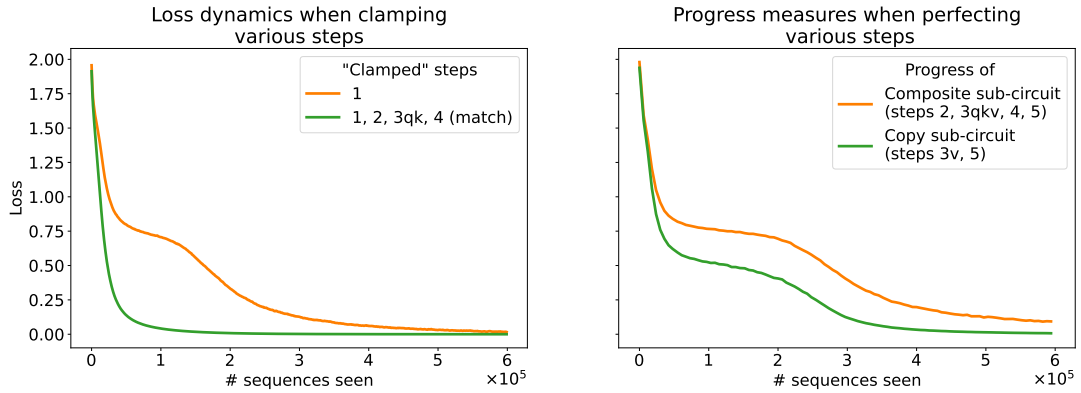


*Figure 10.* (Left) Our clamping method applied to studying formation of composite subcircuit consisting of Steps 2, 3qkv, 4, 5 (orange) and copy subcircuit consisting of Steps 3v, 5 (green). (Right) Progress measures tracking these same circuits. We can see that the clamping analysis more clearly shows the difference in formation dynamics of these two subcircuits, whereas the progress measure shows a phase change for both.

## E. The role of Layer 1 heads

To establish the causal role of Layer 1 heads to individual induction heads in Layer 2, we did extensive ablation analyses.

Earlier, we found that Layer 1 heads do not substantially contribute to the output logits. To see if they only compose to Layer 2 heads' attention patterns (the match subcircuit), we consider a "pattern preserving" ablation (Olsson et al., 2022): the outputs of Layer 1 heads are completely ablated, but the attention patterns in Layer 2 are fixed to what they would've been had the Layer 1 heads not been ablated. We implement this using our artificial optogenetics framework by using two passes through the network. In the first, nothing is ablated. The Layer 2 attention patterns from this pass are then put into the `cache`. On the second pass through the network, Layer 1 outputs are ablated, and the `cache` is used to restore the patterns. We find that accuracy drops substantially (down to 55%), indicating substantial V-composition in our networks. To study V-composition, we introduce an analogous "value-preserving" ablation where instead of caching attention patterns, Layer 2 value vectors are cached. When using just a "value-preserving" ablation of all Layer 1 heads (without preserving patterns), accuracy drops substantially (down to 11%). These results indicate that Layer 1 heads are contributing to both the match subcircuit (patterns) and the copy subcircuit (values). Of the two, the role in the match circuit does seem to be "more important", which aligns with prior work (Olsson et al., 2022).

To verify the observed previous token (PT) heads, we next consider an ablation of these three heads with pattern-preserving (which should have no effect if they are PT heads) or value-preserving (which should have a huge effect, since patterns in Layer 2 will get messed up with these heads deleted). Likewise, we consider both types of ablations of the other heads (Layer 1 heads 0,3,4,6,7). Results are summarized in Table 1. We can clearly see that ablating heads 1,2,5 while preserving patterns has little effect, indicating these heads primarily compose with the match subcircuit. Likewise, we find the other heads primarily participate in V-composition for the copy subcircuit.

*Table 1.* Accuracy after Layer 1 head ablations of various sets of heads. All Layer 2 heads remain active.

| Ablated heads | Type of ablation: | |
|---|---|---|
| | Pattern-preserving | Value-preserving |
| Layer 1 Heads 1,2,5 (PT heads) | 99.85% | 69.87% |
| Layer 1 Heads 0,3,4,6,7 (rest) | 34.27% | 99.32% |

Finally, we wish to establish the connectivity between Layer 1 and Layer 2 heads. To be specific to Layer 2 head, we repeat each experiment 8 times, once for each Layer 2 head active (and the others are ablated). As in Section 3.1, we first consider ablating a single previous token head. Results are shown in Figure 11a. We find that when ablating single heads, only Head 2 seems to have a strong effect. We believe this result is again suffering from the redundancy across heads, so the

importance of heads 1 and 5 is masked. To account for this, we instead consider ablating all but a given Layer 1 head. In this case, each square in the heatmap can be seen as one path through the network, with only one head in each layer used. Results are shown in Figure 11b. From these plots, it's clear that there are multiple active paths in the network:

1. Layer 1 Head 1 $\rightarrow$ Layer 2 Head 3
2. Layer 1 Head 2 $\rightarrow$ Layer 2 Heads 1, 2, 3
3. Layer 1 Head 5 $\rightarrow$ Layer 2 Heads 1 and 3

These results indicate a *many-to-many wiring diagram*.

We'd also like to take a moment here to emphasize the use of activation-level ablation analyses, as opposed to metrics calculated on weights. Composition score (Nanda, 2022; Elhage et al., 2021) is one such weight-based metric, calculated as:

$$\text{score}\left(L_1 H_i \rightarrow L_2 H_j : Q/K/V\right) = \frac{||W_O^{L_1 H_i} W_{Q/K/V}^{L_2 H_j}||_F}{||W_O^{L_1 H_i}||_F ||W_{Q/K/V}^{L_2 H_j}||_F}$$

It aims to measure the connection strength between heads in different layers by considering the matrices they use to write output and read input from the residual stream. In Figure 11c, we plot the composition scores of Layer 1 heads to key and value matrices of Layer 2 heads. Though the most noteworthy cases of composition are recovered (such as the importance of Layer 1 Head 2 for K-composition, and Layer 1 Head 6 for V-composition), overall we find the results to be much harder to interpret than those using our activation-level ablation analyses (Figure 11b).